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Test & Measurement:

Integrating USB 3.0 Into Oscilloscopes Allows GAP-FREE STREAMING

Confused with all the feature updates and new specifications you see on oscilloscopes? This interview takes a look at the broad trends affecting oscilloscopes and how these help engineers. Trevor J. Smith, manager - test and measurement, Pico Technology, speaks with Dilin Anand from EFY

Q. What defines newer oscilloscopes?

A. Newer test equipment and oscilloscopes have much deeper memories. The best among the latest scopes have 512 mega-samples or more of capture or buffer memory.

In 2013, scopes had up to 500MHz with one giga-sample of buffer memory. Today, this has doubled. For example, oscilloscopes with 250MHz to 500MHz bandwidth come with a two giga-sample buffer that can hold two 200m captures at a rate of up to five giga-samples per second.

On the ports side, scopes have now upgraded to use USB 3.0 from USB 2.0 for the interface. While the new interface is obviously faster, the real difference is that, integrating USB 3.0 allows users to stream gap-free continuous data from oscilloscopes onto their computers. They can now graph frequency and duty cycles against time, and capture waveforms in modulation domain rather than frequency and time domain.

Q. What drives the need for additional memory?

A. Test engineers today need to do deep analysis. This creates the need for extremely high sampling speeds, which also increases the need for more memory. Additional memory is also good for analysing strings of data. If design engineers are working on serial standards like I2C and Flexray,

then analysing those communication channels along with collected data would require additional memory.

Q. What about the software side of things?

A. Software upgrades to scopes are a big part of their feature set. Vendors are constantly releasing new versions of software. Sometimes you see software updates deliver support for newer protocols that the original equipment did not ship with.

In January 2016, some oscilloscopes saw feature updates that increased the number of serial decoders to 16 from the seven or eight protocols that these previously had. Sometimes, customers request support for specific protocols like Flexray, and we add those to their test equipment through software updates.

Q. Any changes in the new scopes related to their usage?

A. On the user interface front, it is designed to be more usable for touch based zooming and panning. One interesting feature here is the age counter for people counting long data streams of positive edges; they do not have to manually add these.

Another change is with simplifying visuals. If you look at CAN bus package, there are different fields. What we did was colour-coded these. Now, customers can visually understand the results of the decoder.

Q. Any interesting insights in engineers' preferences for various features or oscilloscopes?

A. While older engineers prefer benchtop instruments, younger ones prefer USB based scopes. Some customers pick up USB based scopes and use it as a component, as an OEM. These guys do not use plain software, but use SDKs to develop their own software. Power analysis, monitoring, medical and other applications are implemented using USB based scopes as a component integrated inside.

Q. What about the broad trends for oscilloscopes as a whole?

A. High bandwidth is definitely a trend. Today, we see oscilloscopes for inspecting cables and transmission lines to look at their characteristics. In a car-wiring loom, we never had to worry about the bandwidth going around the car. Today, if you need a rear-view camera or other high-performance signals, the wiring needs to have high-quality cables.

Traditional loom manufacturers tend to not deliver the quality that is now needed for vehicle manufacturers. As a result we find a lot of need for time-domain reflectometer measurements in automotive. The same happens in robotics and factory automation, where people look at high bandwidth signals. **EFY**